

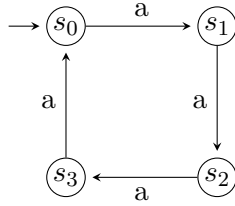
# Concurrent Programming - Exercícios 1

## Labeled Transition Systems

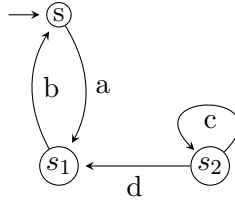
1. What are the values of  $x$  after the execution of the program? How many different executions there are?

$x \leftarrow 10; ((x \leftarrow 2x; x \leftarrow x - 1; x \leftarrow x + 2) \parallel x \leftarrow x - 5)$

2. Consider the following LTS



- (a) Define the LTS as a triple  $(S, \longrightarrow, s_0)$  and determine  $Act$ .
  - (b) Draw the reflexive closure of the binary relation  $\xrightarrow{a}$ .
  - (c) Draw the symmetric closure of the binary relation  $\xrightarrow{a}$ .
  - (d) Draw the transitive closure of the binary relation  $\xrightarrow{a}$ .
3. Let the LTS



- (a) Define the LTS as a triple  $(S, \longrightarrow, s)$  and the set  $Act$ .
  - (b) Compute  $Post(s_1)$  and  $Act(s_2)$
  - (c) Determine  $Reach(s_2)$ .
4. Let  $Post^0(s) = \{s\}$  and  $Post^{n+1}(s) = Post(Post^n(s))$ , show that

$$Reach(s) = \bigcup_n Post^n(s).$$

5. Let  $\longrightarrow^* \subseteq S \times Act^* \times S$  be the reflexive and transitive closure of  $\longrightarrow$ , show that  $s \xrightarrow{\omega}^* s'$  iff  $s' \in Reach(s)$ , where  $\omega = \alpha_1 \cdots \alpha_n$  for some  $\alpha_i \in Act$ . Hint: By induction on  $n$ .
6. For each of the following machines build a LTS that models its behaviour.
  - (a) A machine that given a coin produce coffee
  - (b) A machine that given a coin produce coffee or tea
  - (c) A machine that given a coin one can push a button that allows to choose between coffee or tea
  - (d) A machine as in the previous case but that after producing two beverages stops.

- (e) A machine that given a coin produce coffee but may also not give coffee and return to the initial state

7. Solve the problems of LTSs in PseuCo.com.

8. Two LTS  $TS = (S, \longrightarrow, s_0)$  and  $TS' = (S', \longrightarrow', s'_0)$  are isomorphic,  $TS \sim TS'$ , it there exists a bijection  $f$ ,

$$f : Reach(TS) \rightarrow Reach(TS')$$

with

- $f(s_0) = s'_0$
- for all  $s_1, s_2 \in Reach(TS)$  and for all  $\alpha \in Act$

$$s_1 \xrightarrow{\alpha} s_2 \text{ iff } f(s_1) \xrightarrow{\alpha'} f(s_2)$$

- (a) Show that the LTS isomorphism is a equivalence relation
- (b) Show that a LTS that is finitely branching and that has a finite number of states is isomorphic to a finite-state LTS.